D D M M Y Y Y Y

MODULE - 02.

1] Define process. with a neat diagram, explain the states of a process with a transition diagram and process Control block?

=> A process is a program in execution.
It also includes.
1] program counter to indicate the current activity
2] Registers control of the processor
3] process stack contains temporary data
4] Data section contains global variables
5] Heap is memory that is dynamically allocated during process run time.

PROCESS STATE :-
The state of process is defined in part by the current activity of that process. Each process may be in one of the following state:-

-> New :- The process is being executed.
-> Running :- Instructions are being executed.
-> waiting :- The process is waiting for some event to occur (such as an I/O completion or reception of a signal).
-> Ready : The process is waiting to be assigned to a processor.
-> Terminated :- The process has finished execution.

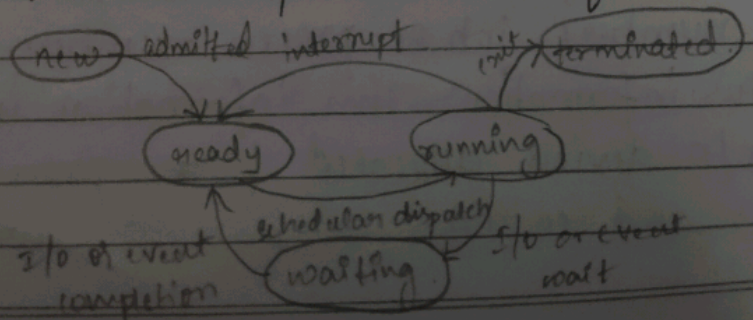


Diagram of process state.

D D M M Y Y Y Y

Process Control Block : (PCB)

- Each process is represented in the OS by a process control block (PCB) - also called a task control block. A PCB is shown in figure.
- It contains many pieces of information associated with a specific process, including these:
1. process state :- The state may be new, ready, running and waiting, halted and so on.
2. program counter: The counter indicates the address of the next instruction to be executed for this process.
3. CPU registers :- They include accumulators, index registers, stack pointers, and general-purpose registers, plus any condition-code information.
4. CPU-scheduling information :- This information includes a process priority, pointers to scheduling queues, and any other scheduling parameters.
5. Memory-management information :- This information may include such information as the values of the base and limit registers, the page tables, or the segment tables, depending on the memory system used by the Operating System.
6. Accounting information :- This information includes the amount of CPU and real time used, time limits, account numbers, job or process numbers, and so on.
7. I/O status information :- This information includes the list of I/O devices allocated to the process, a list of open files and so on.

DD MM YYYY

process state
process number
program counter
registers
memory limits
list of open files
• • •

process control block (PCB).

2] what is inter-process communication? Discuss message passing and the shared memory concept of IPC?

→ process executing concurrently in the OS may be.
1] Independent processes or. 2] Co-operating processes.

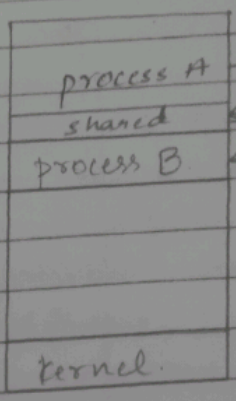
1] A process is independent if.
(i) The process cannot affect or be affected by the other processes.
(ii) The process does not share data with other processes.

2] A process is co-operating if
(i) The process can affect or be affected by the other processes.
(ii) The process shares data with other processes.

Two basic models of IPC.
1) shared memory.
2) Message passing.

D D M M Y Y Y Y

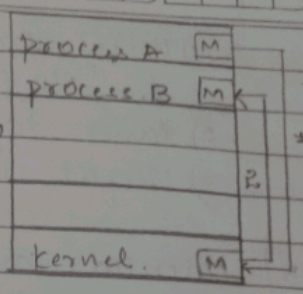
1) shared memory.



- Typically, a shared memory region resides in the address space of the process creating the shared memory segment.
- To illustrate the concept of co-operating processes, let's consider the producer - consumer problem.
- A producer process produces information that is consumed by a consumer ~~problem~~ process.
- One solution to the producer - consumer problem uses shared memory. To allow producer and consumer processes to run concurrently, we must have available a buffer to items that can be filled by the producer and emptied by the consumer.
- This buffer will reside in a region of memory that is shared by the producer and consumer processes. A producer can produce one item while the consumer is consuming another item. so that the consumer does not try to consume an item that has not yet been produced.
- Two types of buffers can be used:-
  1) unBounded - Buffer
  2) BOunded - Buffer.
- Advantages:-
  1) Allow maximum speed & convenience of communication
  2) faster.

D D M M Y Y Y Y

2) Message passing systems:



- MPS provides a mechanism to allow process to communicate and to synchronize their actions without sharing the same address space. and is particularly useful in a distributed environment, where the communicating processes may reside on different computers connected by a network.
- A message-passing facility provides at least two operations:
  1) send (message) and
  2) receive (message)
- messages sent by a process can be of either.
  - fixed or
  - variable size.
- If only fixed-sized messages can be sent, the system-level implementation is straight forward.
- Variable-sized message require a more complex system level implementation. but the programming task becomes simpler.
- A communication link must exist b/w processes to communication.
- Three methods for implementing a link:
  1) Direct or indirect communication.
  2) symmetric or assymmetric communication
  3) Automatic or explicit buffering.
- Two operations:
  1) send (P, message): Send a message to process P.
  2) receive (Q, message): Receive a message from process Q.
- Advantages:

D D M M Y Y Y Y

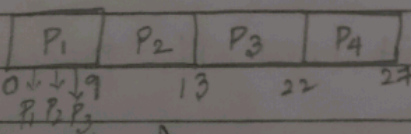
1) Useful for exchanging smaller amounts of data (∵ "No conflicts need beavoided).
2) Easier to implement
3) Useful in a distributed environment.

3] Calculate average waiting and turn around times by drawing the gantt chart using FCFS & RR(q=2ms)

Processes	Arrival time	Burst time
P_1	0	9
P_2	1	4
P_3	2	9
P_4	3	5.

⇒ FCFS :-

P	AT	BT	CT	TAT	WT	RT
P_1	0	9	9	9	0	0
P_2	1	4	13	12	8	8
P_3	2	9	22	20	11	11
P_4	3	5.	27	25	20	20.
				Avg = 16.5	9.75.	



RR (q=2ms).

P	AT	BT	CT	TAT	WT	RT
P_1	0	9	25	25	16	0
P_2	1	4	12	11	7	1
P_3	2	9	26	24	15	2
P_4	3	5.	20	17.	12	5

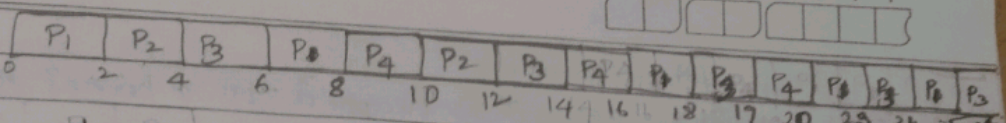
Avg of TAT = 19.25
WT = 12.2

RQ = 7 2 7 5 3 2 5 1 3 3 1 1 1 1

GQ = P1 P2 P3 P1 P4 P2 P3 P4 P1 P3 P4 P1 P3

D D M M Y Y Y Y

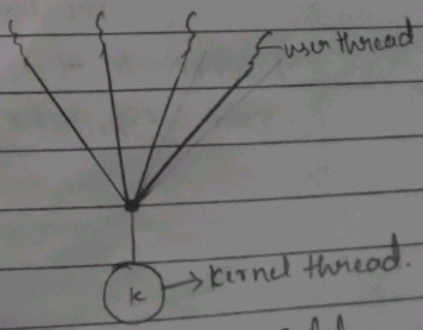
G.C =



4] Discuss in detail the multithreading model, its advantages & disadvantages with suitable illustrations.

⇒ - Support for threads may be provided at either.
1) The User level, for user threads or
2) By the kernel, for kernel threads.
- User-threads are supported above the kernel and are managed without kernel support. Kernel-threads are supported and managed directly by the Os.
- Three ways of establishing relationship b/w user-threads & kernel-threads:

1) Many-to-one model.
- Many user-level threads are mapped to one kernel thread.
- Advantage:-
1) Thread management is done by the thread library in user space, so it is efficient.
- Disadvantages:-
1) The entire process will block if a thread makes a blocking system-call.
2) Multiple threads are unable to run in parallel on multiprocessors.
- Eg:- Solaris green threads
GNU portable threads.



Many-to-one model.

D D M M Y Y Y Y

2) One-to-one model.

- Each user thread is mapped to a kernel thread
- Advantages:-

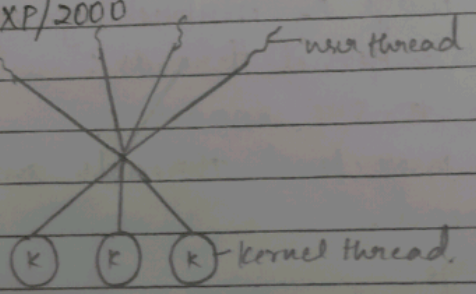
1) It provides more concurrency by allowing another thread to run when a thread makes a blocking system-call.

2) Multiple threads can run in parallel on multiprocessors.

- Disadvantages:-

1) Creating a user thread requires creating the corresponding kernel thread.

- eg:- Windows NT/XP/2000
  Linux.



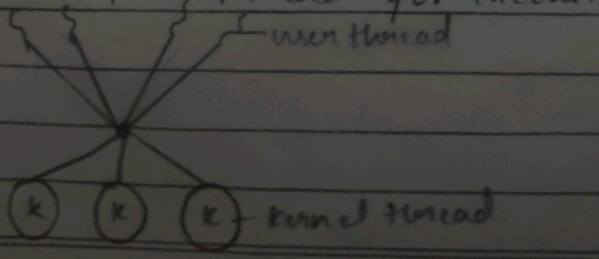
3) Many-to-many Model.

- Many user-level threads are multiplexed to a smaller number of kernel threads
- Advantages:

1) Developers can create as many user threads as necessary.

2) The kernel threads can run in parallel on a multiprocessor.

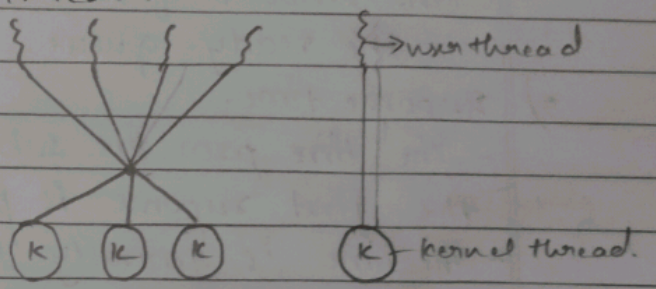
3) When a thread performs a blocking system-call, kernel can schedule another thread for execution.



D D M M Y Y Y Y

Two level model:-

- A variation on the many-to-many model is the two level-model.
- similar to M:M, except that it allows a ~~user~~ user thread to be bound to kernel thread.
- eg:- HP-UX
  Tru64-UNIX



5] Explain five different scheduling criteria used in the computing scheduling mechanism?

→ Different CPU-scheduling algorithms
- have different properties and
- may favor one class to processes over another.

- criteria to compare CPU-scheduling algorithms:

1) CPU Utilization:
- we must keep the CPU as busy as possible.
- In a real system. it ranges from 40% and 90%.

2) Throughput
- Number of processes completed per time unit.
- for long processes, throughput may be 1 process per hour; for short transactions, throughput might be 10 processes per second.

3) Turnaround Time
- The interval from the time of submission of a process to the time of completion.
- Turnaround time is the sum of the periods spent waiting to get into memory.

- waiting in the ready-queue.
- executing on the CPU and
- doing I/O.

4) waiting time.
- The amount of time that a process spends waiting in the ready-queue.

5) Response time.
- The time from the submission of a request until the first response is produced.
- The time is generally limited by the speed of the output device.

* we want
- to maximize CPU utilization & throughput and,
- to minimize turnaround time, waiting time and response time.

6] Explain context-switching?

7] - context-switch means saving the state of the old process and switching the CPU to another process.
- the context of a process is represented in the PCB of the process; it includes.
  - Values of CPU registers
  - process-state and
  - Memory-management information.
- Disadvantage:-

1) context-switch time is pure overhead, because the system does no useful work while switching.
2) context switch times are highly dependent on hardware support.

D D M M Y Y Y Y

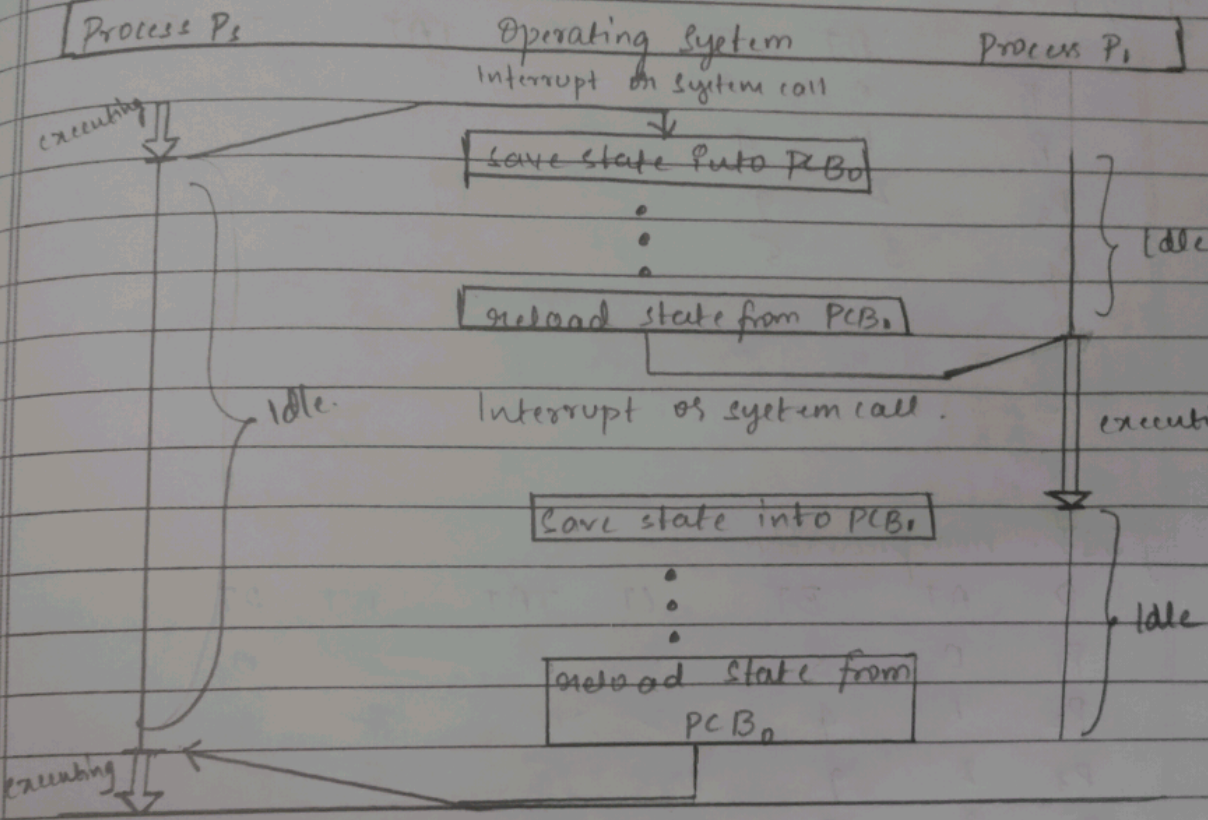


Diagram showing CPU switch from process to process.

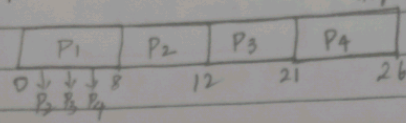
7] Calculate the average waiting time and the average turnaround time by drawing the gantt chart using FCFS, SJF - non preemptive, SRTF, RR (q=2ms) & priority scheduling algorithm?

Process	Arival time	Burst time	Time priority
P_1	0	8	3
P_2	1	4	2
P_3	2	9	1
P_4	3	5	4

D D M M Y Y Y Y

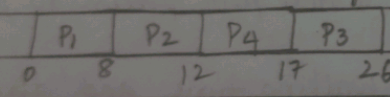
2) 1) FCFS :-

P	AT	BT	CT	TAT	WT	RT
P_1	0	8	8	8	0	0
P_2	1	4	12	11	7	7
P_3	2	9	21	19	10	10
P_4	3	5.	26	23	18	18
				Avg = 15.25	8.75.	



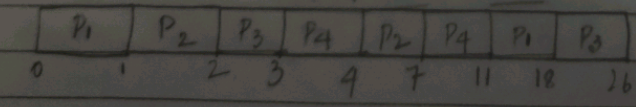
2) SJF - non-preamtive.

P	AT	BT	CT	TAT	WT	RT.
P_1	0	8	8	8	0	0
P_2	1	4	12	11	7	7
P_3	2	9	26	24	15	15
P_4	3	5.	17	14	9	9.
			Avg:	14.25	7.75.	



3) SRTF -> preemptive :-

P	AT	BT	CT	TAT	WT	RT
P_1	0	8 7	18	18	10	10
P_2	1	4 3	7	6	3	3
P_3	2	9 8	26	24	15	15
P_4	3	5 4	11	7	3	3.
			Avg:	13.75	7.75.	



D D M M Y Y Y Y

4] Robin Round Robin wit tq (2ms).

Pn	AT	BT	CT	TAT	WT	RT
P_1	0	8	22	22	14	0
P_2	1	4	18	11	7	1
P_3	2	9	26	24	15	2
P_4	3	5	23	20	15	5

Avg of TAT = 19.25

WT = 12.75

RQ = | P_1 | P_2 | P_3 | P_1 | P_4 | P_2 | P_3 | P_1 | P_4 | P_3 | P_1 | P_4 | P_3 | P_3 |

0

GC = | P_1 | P_2 | P_3 | P_1 | P_4 | P_2 | P_3 | P_1 | P_4 | P_3 | P_1 | P_4 | P_3 | P_3 |

0 2 4 6 8 10 12 14 16 18 20 22 23 25 26

5] priority scheduling

P	AT	BT	CT	TAT	WT	RT
3 – P_1	0	8	8	8	0	0
2 – P_2	1	4	21	20	16	16
1 – P_3	2	9	17	15	6	6
4 – P_4	3	5	26	23	18	18

Avg of TAT = 16.5

WT = 10

P_1	P_3	P_2	P_4

0 8 17 21 26